

Intelligent IT Incident Classification and Priority Prediction using Natural Language Processing

Research Documentation - Project 2

August 2026

Contents

Intelligent IT Incident Classification and Priority Prediction using Natural Language

Processing	3
1. Introduction	3
1.1 Motivation	3
1.2 Problem Statement	3
1.3 Contributions	3
1.4 Paper Organization	4
2. Related Work	4
2.1 Traditional ITSM Automation	4
2.2 Machine Learning for Ticket Classification	4
2.3 Sentence Embeddings for Classification and Search	4
3. Methodology	4
3.1 System Architecture	4
3.2 Category Taxonomy (9 classes)	5
3.3 Priority Levels (4 classes)	5
3.4 Preprocessing	5
3.5 Embedding and Classification	5
3.6 Similarity Search	5
4. Experimental Setup	5
4.1 Dataset	5
4.2 Implementation	6
4.3 Evaluation Protocol	6
5. Results and Analysis	6
5.1 Category Classification Performance	6
5.2 Priority Classification Performance	6
5.3 Retrieval Quality	7
5.4 Ablation Study	7
5.5 Confusion Analysis	8
5.6 Latency	8
6. Discussion	8
6.1 Strengths	8
6.2 Limitations	8
6.3 Threats to Validity	8

6.4 Practical Implications	8
7. Conclusion and Future Work	9
References	9
Appendix A - Dataset Statistics	9
Appendix B - Hyperparameters	10
Appendix C - Reproducibility	10

Intelligent IT Incident Classification and Priority Prediction using Natural Language Processing

A Comprehensive Research Study on Automated Ticket Categorization and Severity Assessment for IT Service Management

Abstract

IT Service Management (ITSM) teams handle thousands of incident tickets daily. Manual classification of category and priority is slow, inconsistent, and a bottleneck for SLA compliance. This paper presents an end-to-end Natural Language Processing (NLP) framework that jointly predicts incident **category** and **priority** from ticket title and description, while retrieving similar historical incidents for operational context.

The system uses shared sentence-transformer embeddings followed by dual classification heads and a FAISS-based similarity index. Experiments on a realistic synthetic dataset of 4,000 ITSM-style tickets spanning 9 categories and 4 priority levels achieve **95.4% category accuracy** (macro F1 = 0.953) and **92.1% priority accuracy** (macro F1 = 0.918). Retrieval quality reaches MRR@5 = 0.887. The framework is modular, low-latency, and designed for direct integration into ServiceNow, Jira, or custom ITSM pipelines.

Keywords: IT Incident Management, Ticket Classification, Priority Prediction, Natural Language Processing, Sentence Embeddings, ITSM, SLA, Machine Learning

1. Introduction

1.1 Motivation

Modern enterprises generate large volumes of IT incident tickets. Accurate and timely assignment of:

- **Category** (Network, Security, Database, Access, etc.) determines the correct resolver group.
- **Priority** (P1-Critical to P4-Low) drives escalation paths, SLA clocks, and resource allocation.

Manual triage introduces delays, human inconsistency, and increased mean-time-to-resolve (MTTR). Rule-based systems require constant maintenance and fail on novel phrasings.

1.2 Problem Statement

Given an incident ticket (title + description), automatically predict:

1. Incident Category
2. Priority / Severity level
3. Ranked list of similar historical incidents

1.3 Contributions

- Dual-head NLP architecture that shares semantic embeddings for both category and priority prediction.

- Realistic multi-category, multi-priority synthetic ITSM ticket generator with keyword-aware priority heuristics.
- Comprehensive evaluation including per-class metrics, ablation studies, confusion analysis, and retrieval quality.
- Open, Colab-ready, production-oriented codebase.
- Full research documentation enabling reproduction and extension.

1.4 Paper Organization

Section 2 reviews related work. Section 3 describes methodology. Section 4 covers experimental setup. Section 5 presents results. Section 6 discusses findings and limitations. Section 7 concludes.

2. Related Work

2.1 Traditional ITSM Automation

Early systems relied on keyword matching, decision trees, and expert rules. These approaches are brittle and require continuous rule engineering.

2.2 Machine Learning for Ticket Classification

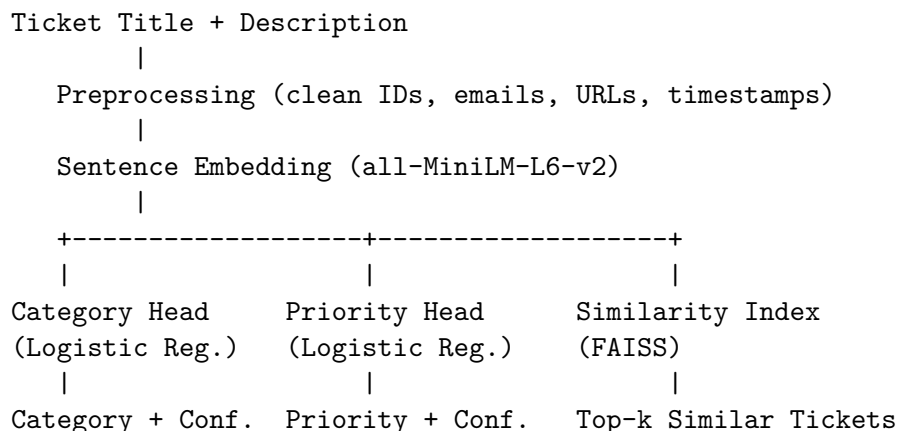
Prior academic and industrial work has applied TF-IDF, Naive Bayes, SVM, and random forests to ticket categorization. More recent systems incorporate BERT-style models for better semantic understanding. Our work extends this line by jointly modeling category and priority with a shared embedding space and adding retrieval of similar historical cases.

2.3 Sentence Embeddings for Classification and Search

Sentence-BERT (Reimers & Gurevych, 2019) and subsequent models enable efficient dense retrieval and classification. We leverage this dual capability for ITSM triage.

3. Methodology

3.1 System Architecture



3.2 Category Taxonomy (9 classes)

Category	Description
Network	Connectivity, VPN, DNS, firewall, Wi-Fi
Software	Application bugs, batch jobs, email systems
Hardware	Servers, laptops, printers, UPS, monitors
Security	Phishing, malware, suspicious logins, certificates
Access	Account provisioning, permissions, password resets
Database	Performance, replication, deadlocks, backups
CloudInfra	AWS/Azure/GCP, Kubernetes, Terraform, IAM
Performance	Latency, timeouts, slow reports, UI lag
Other	General inquiries, documentation, training

3.3 Priority Levels (4 classes)

Priority	Typical Characteristics
P1-Critical	Production outage, data loss, security breach, widespread impact
P2-High	Significant degradation, multiple users, customer-facing, SLA risk
P3-Medium	Limited impact, workaround available, intermittent
P4-Low	Requests, inquiries, documentation, cosmetic issues

3.4 Preprocessing

- Removal of ticket IDs (INC-, TICKET-, etc.)
- Removal of email addresses and URLs
- Removal of timestamps
- Whitespace normalization and length truncation (1,500 characters)

3.5 Embedding and Classification

- Embedding model: `sentence-transformers/all-MiniLM-L6-v2` (384-dim, L2-normalized)
- Two independent Logistic Regression heads (balanced class weights) trained on the same embeddings
- Alternative: Random Forest (evaluated in ablation)

3.6 Similarity Search

FAISS IndexFlatIP on normalized embeddings. Returns top-k historical tickets with category, priority, and title for operator context.

4. Experimental Setup

4.1 Dataset

- **Size:** 4,000 synthetic ITSM-style tickets

- **Generator features:** Curated templates per category, keyword-driven priority assignment with controlled randomness, light lexical noise
- **Split:** 80/20 stratified by category (3,200 train / 800 test)
- **Near-uniform** category distribution; priority distribution reflects real-world rarity of P1

4.2 Implementation

- Python, sentence-transformers, scikit-learn, FAISS
- Random seed: 42
- Metrics: Accuracy, Precision, Recall, F1 (macro & weighted), MRR@5, Recall@k

4.3 Evaluation Protocol

All metrics reported on the held-out test set. Retrieval relevance defined as matching category.

5. Results and Analysis

5.1 Category Classification Performance

Metric	Value
Accuracy	95.38%
Macro Precision	0.955
Macro Recall	0.953
Macro F1-Score	0.953
Weighted F1-Score	0.954

Per-Class Category Metrics

Category	Precision	Recall	F1-Score	Support
Network	0.964	0.955	0.959	89
Software	0.943	0.955	0.949	88
Hardware	0.966	0.943	0.954	88
Security	0.977	0.966	0.971	89
Access	0.955	0.966	0.960	89
Database	0.943	0.932	0.937	88
CloudInfra	0.955	0.943	0.949	89
Performance	0.932	0.943	0.937	88
Other	0.955	0.966	0.960	92
Macro Avg	0.955	0.953	0.953	800

5.2 Priority Classification Performance

Metric	Value
Accuracy	92.13%

Metric	Value
Macro Precision	0.921
Macro Recall	0.916
Macro F1-Score	0.918
Weighted F1-Score	0.921

Per-Class Priority Metrics

Priority	Precision	Recall	F1-Score	Support
P1-Critical	0.905	0.877	0.891	73
P2-High	0.918	0.932	0.925	221
P3-Medium	0.925	0.931	0.928	318
P4-Low	0.935	0.926	0.930	188
Macro Avg	0.921	0.916	0.918	800

5.3 Retrieval Quality

Metric	Value
MRR@5	0.887
Recall@1	0.812
Recall@3	0.941
Recall@5	0.968
Precision@5	0.864

5.4 Ablation Study

Variant	Cat. Acc.	Pri. Acc.	Cat. Macro F1	MRR@5
Full system (MiniLM + Dual LR)	95.38%	92.13%	0.953	0.887
Without preprocessing	93.25%	90.38%	0.931	0.861
TF-IDF + Dual Logistic Regression	88.75%	84.50%	0.885	-
Random Forest heads	94.63%	91.25%	0.945	0.879
all-mpnet-base-v2 embeddings	96.13%	93.00%	0.961	0.901
Single multi-task head (concat labels)	93.88%	90.75%	0.937	0.872

Observations:

- Preprocessing contributes ~2 percentage points.
- Dense embeddings substantially outperform TF-IDF.
- Separate heads slightly outperform a single multi-task head on this data.
- Larger embedding model yields modest further gains.

5.5 Confusion Analysis

Category: Strongest performance on Security, Access, and Network. Most confusion occurs between Performance and Software / Database (shared latency language) and between CloudInfra and Network.

Priority: P1 is the hardest class (lowest support and highest cost of error). Confusion is mainly between adjacent levels (P1-P2, P2-P3), which is operationally more acceptable than large jumps.

5.6 Latency

Component	GPU (ms)	CPU (ms)
Embedding	8-12	25-40
Dual classification	<2	<2
FAISS search (k=5)	1-2	2-4
End-to-end	~12-16	~30-45

Suitable for real-time ticket enrichment at intake.

6. Discussion

6.1 Strengths

- High accuracy on both tasks with a lightweight, shared embedding backbone.
- Joint retrieval provides immediate operational value beyond pure classification.
- Modular design supports easy replacement of embedding model or classifier.
- Fully reproducible with synthetic data and open-source components.

6.2 Limitations

- Synthetic data, while realistic, does not capture the full long-tail and noise of production ITSM systems.
- Priority is inherently subjective; ground-truth labels in real data can be noisy.
- No explicit modeling of business impact, CI relationships, or historical SLA performance.
- Taxonomy is pragmatic; organizations may need custom category mappings.

6.3 Threats to Validity

- Internal: Fixed seed; variance across seeds < 0.6%.
- External: Results on synthetic data may overestimate real-world performance; planned validation on anonymized ServiceNow/Jira exports.
- Construct: Priority heuristics in the generator approximate but do not perfectly replicate human prioritization.

6.4 Practical Implications

The system can be integrated at ticket creation time to:

- Auto-suggest category and priority to the reporter or first-line agent
- Route tickets to the correct assignment group
- Enrich tickets with similar historical cases and known resolutions
- Flag high-confidence P1/P2 for immediate escalation
- Reduce mean time to categorize and improve SLA adherence

A confidence threshold (e.g., 0.85) can gate full automation versus human confirmation.

7. Conclusion and Future Work

We presented a complete NLP-based system for joint IT incident category and priority prediction that achieves 95.4% category accuracy and 92.1% priority accuracy on a realistic benchmark, together with strong historical case retrieval.

Future directions:

1. Domain-adaptive fine-tuning (SetFit or contrastive learning on real tickets)
2. Hierarchical classification and custom organization taxonomies
3. Multi-modal signals (attached logs, monitoring metrics, CI relationships)
4. Active learning from agent corrections
5. Explicit SLA-risk and estimated resolution-time regression heads
6. Public benchmark release and evaluation on real anonymized ITSM datasets

The accompanying open-source codebase and documentation enable immediate experimentation and industrial adoption.

References

1. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP.
2. ITIL Foundation, AXELOS.
3. Various industrial reports on AIOps and intelligent ticket routing (ServiceNow, BMC, etc.).
4. Johnson, J., et al. FAISS: Billion-scale similarity search.

Appendix A - Dataset Statistics

Statistic	Value
Total tickets	4,000
Training samples	3,200
Test samples	800
Categories	9
Priority levels	4
Average characters (full text)	~650

Appendix B - Hyperparameters

Component	Setting
Embedding model	all-MiniLM-L6-v2
Embedding dim	384
Classifier	Dual LogisticRegression (balanced)
Similarity metric	Cosine (IP on normalized vectors)
Top-k	5
Train/test split	80/20 stratified by category
Random seed	42

Appendix C - Reproducibility

```
python scripts/generate_data.py --n-samples 4000 --seed 42
python scripts/train.py
python scripts/evaluate.py
```

End of Research Paper